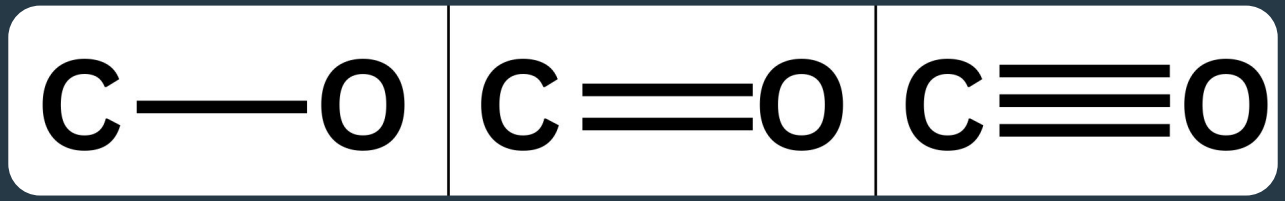
Computational Exploration of Bond Order Calculations Through Density Matrices

•••

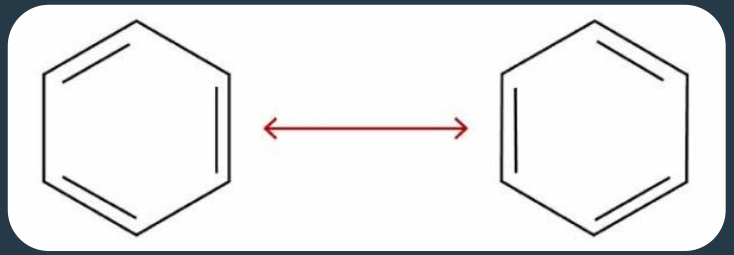
CHEM 279

Lily Hirano & Snigdha Chanduri

Background

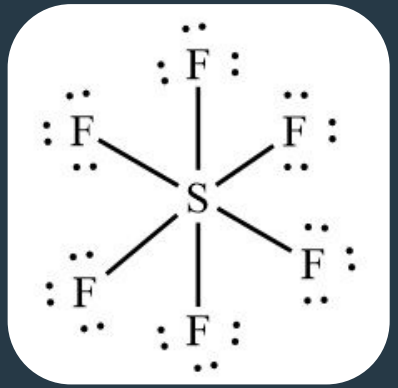


- Bond order quantifies the number of chemical bonds between atoms
- Important for:
  - Bond formation availability
  - Stability
  - Reactivity
  - Classification & modeling



benzene

- No universal definition for bond order currently exists
  - Bonds are more complex than Lewis structures
  - How much electron density is shared between two atoms
- Formulations need to be:
  - Dynamic
  - Influenced by energy and bond length



sulfur hexafluoride

Investigation into how different formulations describe electron sharing and chemical bonding in molecular systems

Methodology

1. Created C++ program capable of
   a. performing SCF calculations
   b. generating molecular overlap and density matrices

2. Implemented three different bond order calculation methods

3. Compare the mathematical formulations and computational behavior of the approaches

4. Determine how closely the calculated values align with expected chemical behavior

5. Evaluate how each bond order method represents bonding interactions across different molecular systems

Three different approaches for bond order

- Each formulation has its advantages and disadvantages
- Some extend Lewis structures by showing other electric interactions
- All use the density matrix to approximate bond order

Wiberg

- For orthogonal basis sets
- Uses density matrix only

$$B_{AB} = \sum_{\mu \in A} \sum_{\nu \in B} P_{\mu\nu}^2$$

Mayer

- Can include non-orthogonal basis sets
- Includes overlap matrix

$$B_{AB} = \sum_{\mu \in A} \sum_{\nu \in B} PS_{\mu\nu} PS_{\nu\mu}$$

Mulliken

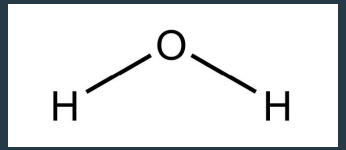
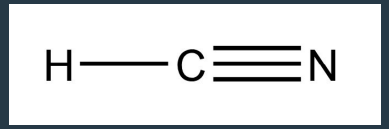
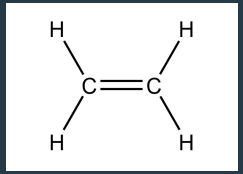
- Splits electrons between atoms 50/50
- Multiplied by 2 for each spin

$$B_{AB} = 2 \sum_{\mu \in A} \sum_{\nu \in B} P_{\mu\nu} S_{\mu\nu}$$

Molecule types

Name	Structure	Type
hydrogen	H—H	simple diatomic
hydrogen fluoride	H—F	polar diatomic
hydroxyl	H—O	radical

Molecule types

Name	Structure	Type
water		polar polyatomic
hydrogen cyanide		linear triple-bond
ethene		planar double-bond

Results & Analysis

Molecule	Mean Difference		
	Wiberg	Mayer	Mulliken
H_2	4.44×10^{-16}	0.378	0.320
HF	0.050	0.148	0.009
HO	0.025	0.235	0.084
H_2O	0.013	0.195	0.069
HCN	0.055	1.11	0.787
C_2H_4	0.346	0.725	0.362

Direct comparison of molecules – H2O

Wiberg	O	H1	H2
O	0	0.981	0.981
H1	0.981	0	1e-5
H2	0.981	1e-5	0

Mayer	O	H1	H2
O	0	1.26	1.26
H1	1.26	0	0.057
H2	1.26	0.057	0

- Wiberg → how many bonds are occuring
- Larger Mayer value → stronger electron sharing / covalency
- Mulliken → simpler partitioning

Mulliken	O	H1	H2
O	0	1.10	1.10
H1	1.10	0	0.002
H2	1.10	0.002	0

Direct comparison of molecules – HF

Wiberg	H	F
H	0	0.950
F	0.950	0

Mayer	H	F
H	0	1.15
F	1.15	0

- Mayer shows
  - Polarized bond
  - Substantial overlap
- Mulliken
  - Best suited for highly polar bonds

Mulliken	H	F
H	0	0.991
F	0.991	0

Direct comparison of molecules – HCN

Wiberg	C	N	H
C	0	3.04	0.936
N	3.03	0	0.061
H	0.936	0.061	0

Mayer	C	N	H
C	0	5.01	1.89
N	5.01	0	0.447
H	1.89	0.447	0

- Wiberg → calculated on orthogonal basis
- Mayer → high overlap increases value
- Mulliken → non equal split of shared electrons

Mulliken	C	N	H
C	0	4.65	1.68
N	4.65	0	0.030
H	1.68	0.030	0

No equation has the ‘best’ bond order

- Each formulation has its own use cases
- Wiberg
  - Generally useful for quantifying bond orders closest to simple integer values
  - Not useful for highly polar molecules
- Mulliken
  - Useful for quantifying polar bond orders closest to simple integer values
- Mayer
  - Added definition of the bond strength and stability
  - More useful for complex molecules that are non-orthogonal
  - More computationally intense

Thanks!